



Bridging the Black Box: Explainable Anomaly Detection for Critical Infrastructure

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Abstract

With the increasing use of computer systems in critical infrastructure, the need to secure and protect them has grown exponentially. Cyberattacks have become increasingly more common and complex, which further increases the chance of a malicious actor targeting our infrastructure. Deep Learning Models have shown promise of being effective in detecting anomalies but are known for their “black box” nature.

Our CIS Framework looks to address the black box nature with:

- Explainable AI Frameworks
- Knowledge Graphs
- Cybersecurity RAG Knowledge Base

Background

What is Critical Infrastructure?

Critical infrastructure systems (smart grid, water, gas, communication, etc.) are ones deemed essential as their failure or disruption would have severe consequences for public safety and national security.

The Dataset

- Data from Dataset [1] which simulates a water distribution system.
- Network data from 1 HMI, 4 PCL, 2 Flow Sensors.
- Physical data including sensor readings and actuator states



Fig 1. Testbed from Dataset
Source: [1]

How are we representing Critical Infrastructure?

- **System Graph** that expresses the testbed system's water flow and network communication.
- **Snapshot Graph** that expresses aggregate values for network traffic and physical measurements of the system.

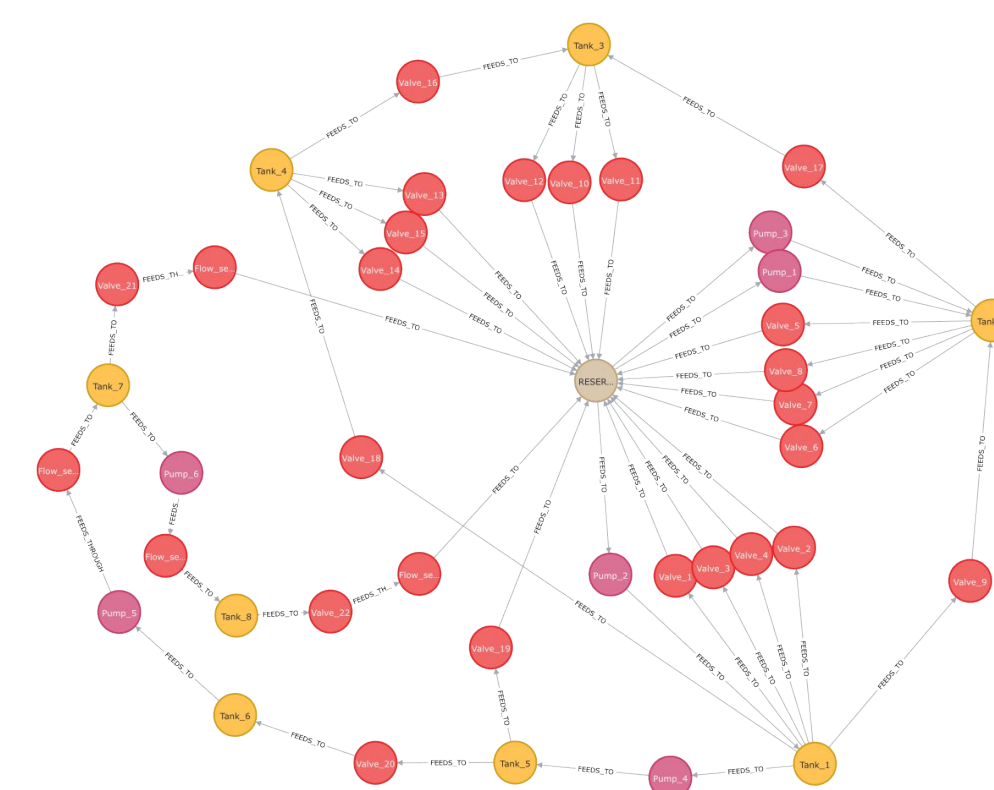


Fig 2. System Knowledge Graph

What types of anomalies are we detecting?

Attack Type	Attacks
Cyber	Denial of Service, scanning, MITM attacks
Physical	Water Leak, Pump Failure, Valve Breakdown

Methods

Data Ingestion

- Network and Physical data loaded into PostgreSQL tables
- Network and Physical data aggregated into 30 second Buckets.
 - Adds averages, mins, max, and counts Features
- Each time bucket is translated into a Snapshot Graph
 - **Nodes:** Endpoint, Connection, Asset, Measurement
 - **Relationships:**
 - (:Snapshot)-[:Contains]->(:Measurement|Connection)
 - (:Endpoint)-[:Initiates]->(:Connection)
 - (:Connection)-[:Terminates_At]->(:Endpoint)
 - (:Asset)-[:Has_Measurement]->(:Measurement)
 - (:Asset)-[:Has_Endpoint]->(:Endpoint)
 - (:Asset)-[:Controls]->(:Asset)
 - (:Asset)-[:Feeds_To|Feeds_Through]->(:Asset)
 - (:Asset)-[:Issues_Command_To]->(:Asset)

GNN Anomaly Detection Model

- Heterogeneous Data allows different node types and Dimensions
- Heterogeneous Graph Transformer (HGTCnv) Layers
- Used for Anomaly Detection

XGBoost Anomaly Classification Model

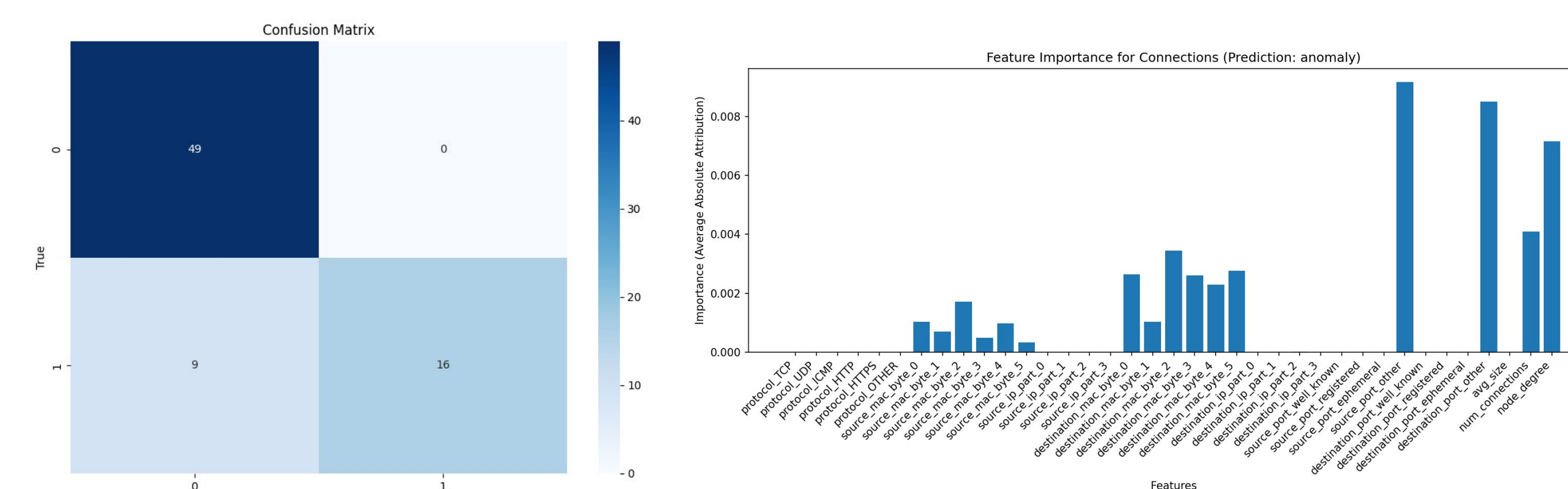
- Uses Combined Aggregated Physical and Network data
- 600 Estimators with Max Depth of 8
- logloss Evaluation Metric

Explainability Methods

- Graph level XAI using Captum Explainer
- Node Importance calculated using Absolute Sum of Attribution
- Feature Importance calculated using Absolute Mean of Attribution

Results

	Accuracy	Recall	Precision	F1 Score
GNN Anomaly Detection	0.88	0.64	1.00	0.78
XGBoost Anomaly Classifier	0.99	0.98	0.98	0.98



Framework Components

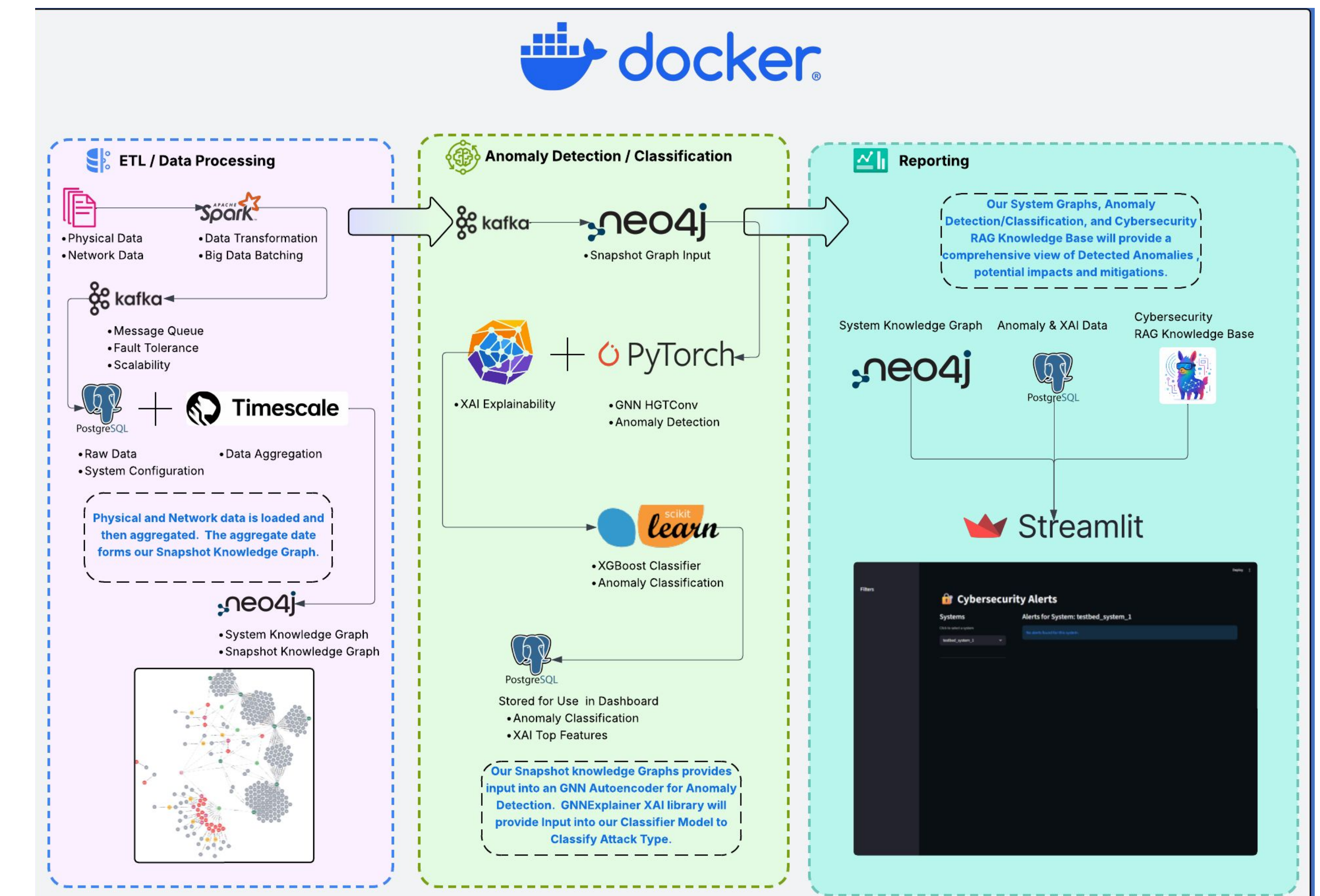


Fig 4. Framework Process Flow

Future Works

Development of our framework is working towards a minimum viable product (MVP) framework available in the coming weeks with the following milestones:

- **Integrate XGBoost Anomaly Classification Model into Anomaly Detection pipeline.**
- **Map explainability nodes to System Knowledge graph node**
- **Optimize data cleaning for GNN training and predictions**
- **Build a Cybersecurity Knowledge Base using RAG-powered LLM.**
- **Build a Cybersecurity Alerts Dashboard to visualize detected anomalies and additional context provided by XAI and Knowledge Base.**

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References

1. L. Faramondi, et al, "A Hardware-in-the-Loop Water Distribution Testbed Dataset for Cyber-Physical Security Testing," doi: 10.1109/ACCESS.2021.3109465.

Dataset

1. Simone Guarino, Luca Faramondi, Roberto Setola, Francesco Flammini (2021). A hardware-in-the-loop water distribution testbed (WDT) dataset for cyber-physical security testing. IEEE Dataport. <https://dx.doi.org/10.21227/rbvf-2h90>